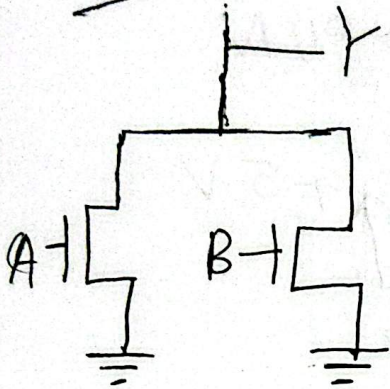
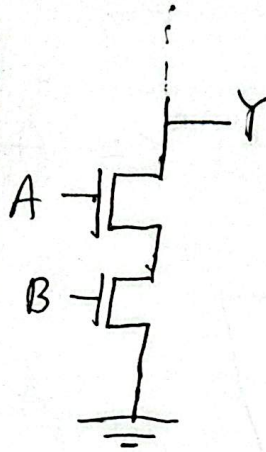


EEE
PDN°



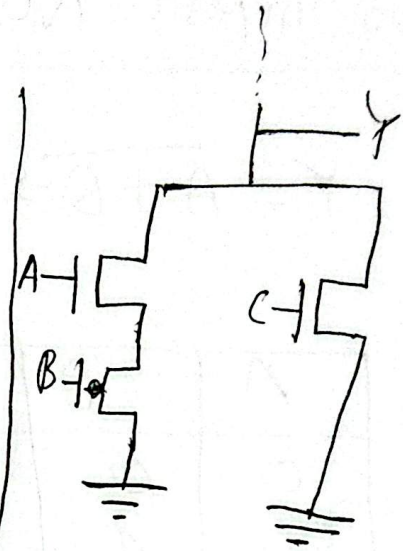
$$Y = \overline{(A + B)}$$

$$\Rightarrow \bar{Y} = A + B$$



$$\bar{Y} = A \cdot B$$

$$\Rightarrow Y = \overline{A \cdot B}$$

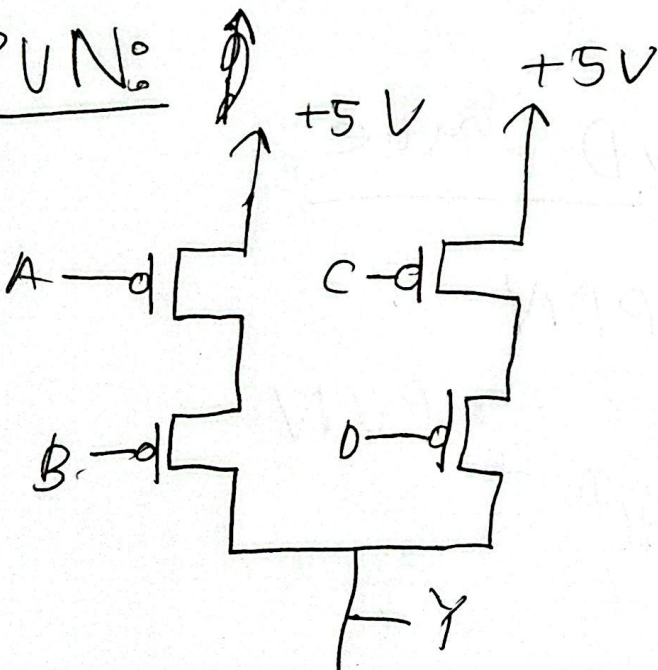


$$\bar{Y} = A \cdot B + C$$

$$Y = \overline{A \cdot B + C}$$

~~PDN~~

PUN°

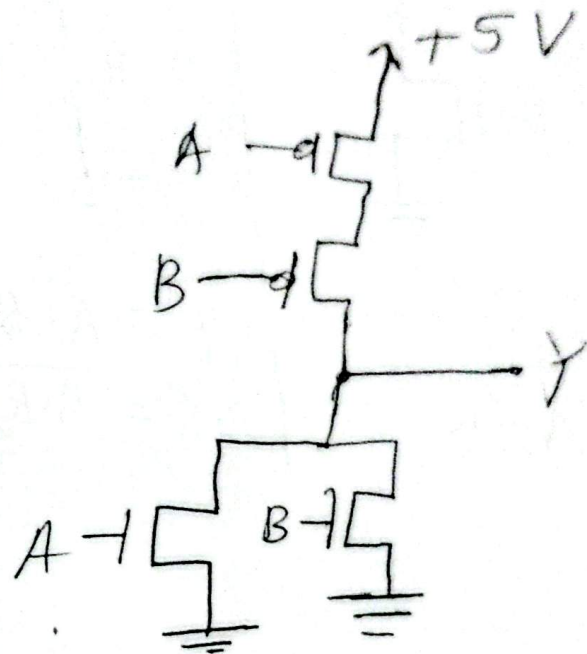


$$Y = \bar{A} \bar{B} + \bar{C} \bar{D}$$

two input NOR gate

$$Y = \overline{A+B} \rightarrow \text{PDN}; \quad Y = \bar{A} \cdot \bar{B} \rightarrow \text{PUN}$$

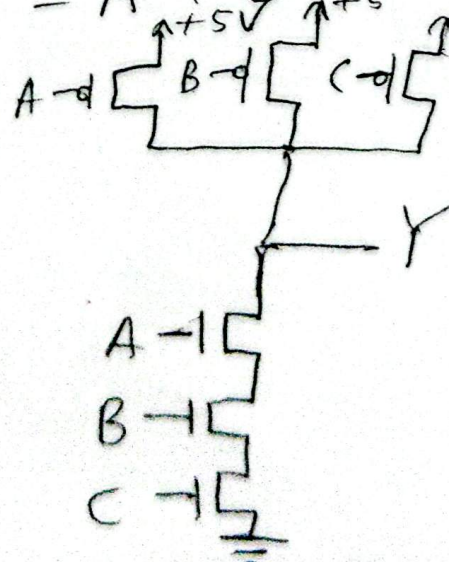
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



three input NAND gate:

$$Y = \overline{A \cdot B \cdot C} \rightarrow \text{PDN}$$

$$\Rightarrow Y = \bar{A} + \bar{B} + \bar{C} \rightarrow \text{PUN}$$

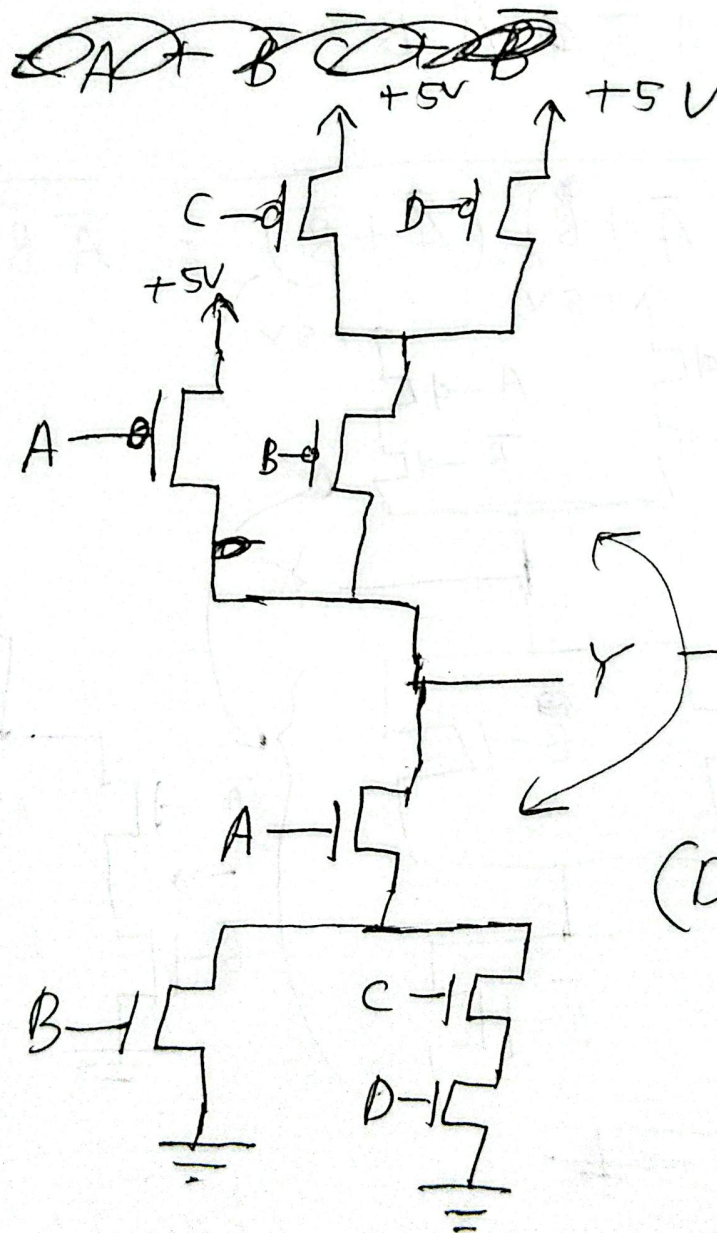


$$Y = \overline{A(B + CD)}$$

$$Y = \bar{A} + \overline{(B + CD)}$$

$$= \bar{A} + \bar{B} \cdot \bar{C} \bar{D}$$

$$= \bar{A} + \bar{B}(\bar{C} + \bar{D})$$



change
all series
with parallel.
(Duality
property)

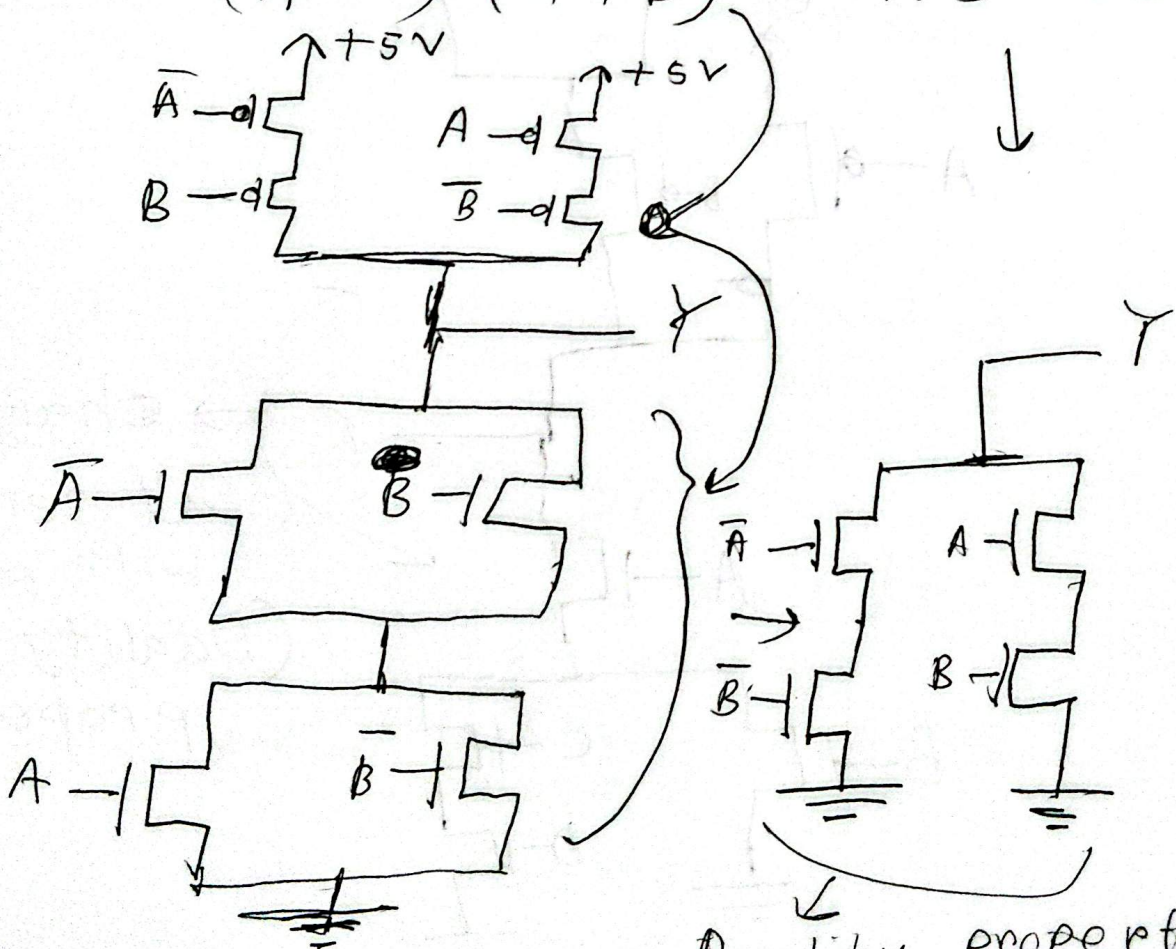
XOR Gate:

$$Y = A\bar{B} + \bar{A}B = \overline{\bar{A}\bar{B}} + \overline{\bar{A}\bar{B}}$$

$$= \overline{A\bar{B} + \bar{A}B}$$

$$= \overline{A\bar{B}} \cdot \overline{\bar{A}B}$$

$$= (\bar{A} + B)(A + \bar{B}) = \bar{A}\bar{B} + AB$$



Duality property is not followed here.

Assignment:

$$Y = \overline{A + B(C + D)}$$

Build the CN